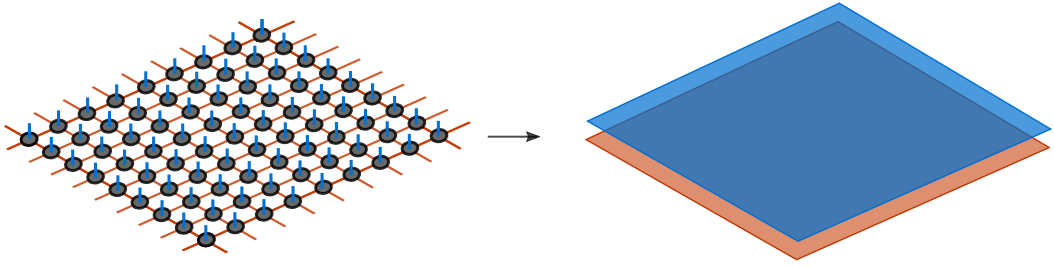
Continuous Tensor Network States of Quantum Fields

Antoine Tilloy, with J. Ignacio Cirac
Max Planck Institute of Quantum Optics, Garching, Germany

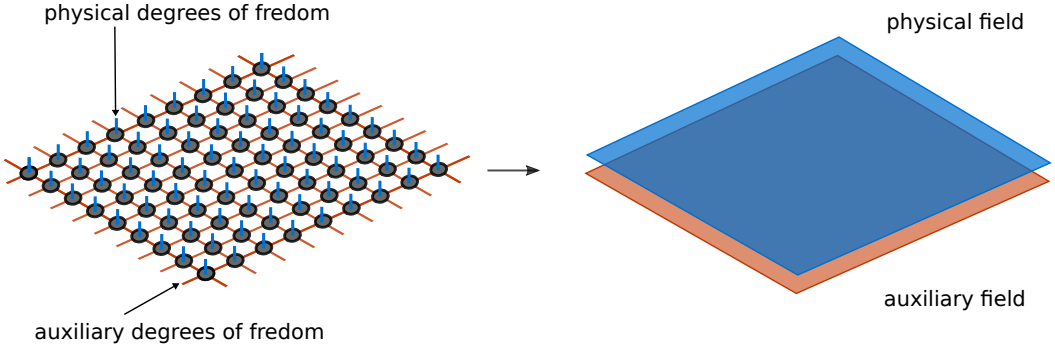


Tensor Networks and Entanglement
Galileo Galilei Institute, Florence
June 18th, 2018

Alexander von Humboldt
Stiftung/Foundation

MPQ

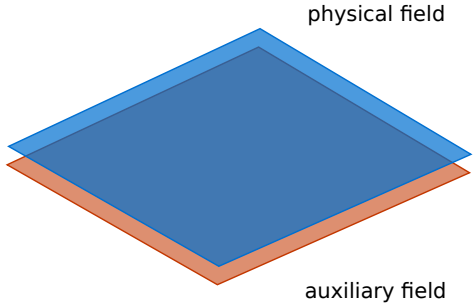
Objective


physical degrees of fredom
auxiliary degrees of fredom
physical field
auxiliary field

Objective

Why?

- **Computations**: the continuum brings new methods (perturbative expansions, saddle point approximations, differential equations)
- **QFT**: apply directly to QFT, without discretization
- **Symmetries**: Implement Euclidean / Translation invariance exactly



Introduction: Matrix Product States

Many-body states are complicated.

$$|\psi\rangle = \sum_{i_1, i_2, \cdots, i_n} c_{i_1, i_2, \cdots, i_n} \, |i_1, \cdots, i_n\rangle$$

2^n parameters $c_{i_1, i_2, \cdots, i_n}$.

Typical many-body Hamiltonians are simple:

$$H = \sum_{k=1}^{n} h_k$$

$\sim$ const $\times$ n parameters.

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Matrix Product States (MPS)

$$|A, L, R\rangle = \sum_{i_1, i_2, \cdots, i_n} \langle \mathrm{L} | A_{i_1}(1) A_{i_2} \cdots A_{i_n}(n) | \mathrm{R} \rangle \, |i_1, \cdots, i_n\rangle$$

A_i are $D \times D$ matrices, hence A is a $2 \times D \times D$ tensor. $|L\rangle$ and $|R\rangle$ are D-vectors.

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The **notation**: $[A_i]_{\mathbf{k},\mathbf{l}} =$ and  $= \sum \delta_{k,l}$ gives:

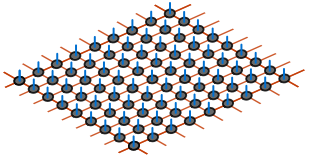
$|A,L,R\rangle =$ 

Introduction: other tensor network states

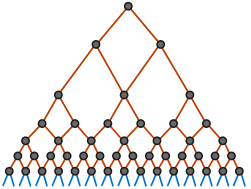
Matrix Product States (MPS)



Projected Entangled Pair States (PEPS)

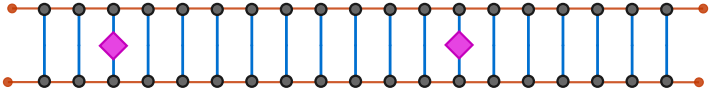


Multi-scale Entanglement Renormalization Ansatz (MERA)



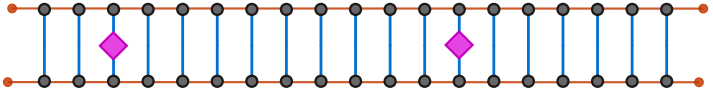
Matrix Product States

Computing expactation values of local observables:

$\langle A|\mathcal{O}(i_k)\mathcal{O}(i_\ell)|A\rangle =$ 

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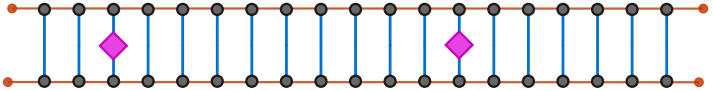
can be done by iteration 2 maps:

$\Phi =$  and $\Phi_{\mathcal{O}} =$ 

i.e. $\langle A|\mathcal{O}(i_k)\mathcal{O}(i_\ell)|A\rangle = \mathrm{tr}(|L\rangle\langle L|\ \Phi^k \cdot \Phi_{\mathcal{O}} \cdot \Phi^{\ell-k} \cdot \Phi_{\mathcal{O}} \cdot \Phi^{n-\ell}\ |R\rangle\langle R|)$

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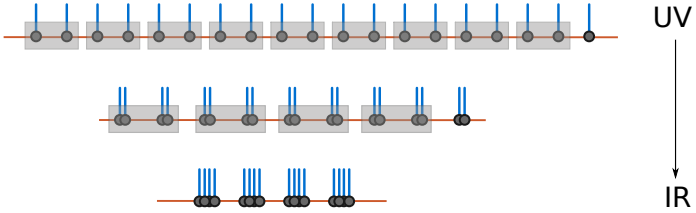
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The contraction for a $d = 1$ system, can be seen as an open-system dynamics in $d = 0$.

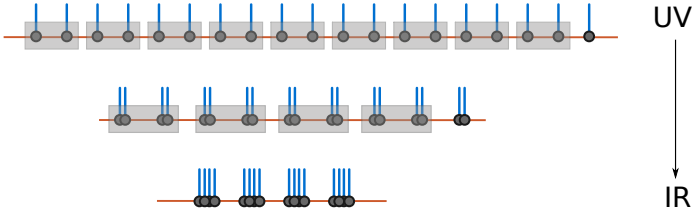
Continuous Matrix Product States

Taking the continuum limit of a MPS



Continuous Matrix Product States

Taking the continuum limit of a MPS



- the bond dimension stays fixed
- the local physical dimension explodes $\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 \longrightarrow \mathscr{F}(L^2([x, x+dx]))$.
  $\Longrightarrow$ **Spins** become **fields**.

Continuous Matrix Product States

Type of ansatz

- Matrices $A_{i_k}(x)$ where the index i_k corresponds to $\psi^{\dagger i_k}(x)|0\rangle$ in physical space.

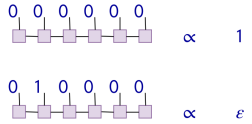
Informal cMPS definition

$$A_0 = \mathbb{1} + \varepsilon Q$$
$$A_1 = \varepsilon R$$
$$A_2 = \frac{(\varepsilon R)^2}{\sqrt{2}}$$
$$\dots$$
$$A_n = \frac{(\varepsilon R)^n}{\sqrt{n}}$$
$$\dots$$

so we go from ∞ to 2 matrices

Fixed by:

- Finite particle number



- Consistency



Continuous Matrix Product States

Definition

$$|Q, R, \omega\rangle = \langle \omega_L | \mathcal{P} \exp\left\{ \int_0^L dx \; Q \otimes \mathbb{1} + R \otimes \psi^\dagger(x) \right\} |\omega_R\rangle \; |0\rangle$$

- Q, R are $D \times D$ matrices,
- $|\omega_L\rangle$ and $|\omega_R\rangle$ are boundary vectors $\in \mathbb{C}^D$,
- $[\psi(x), \psi^\dagger(y)] = \delta(x - y)$

Idea:

$$\begin{aligned} A(x) &\simeq A_0 \mathbb{1} + A_1 \psi^\dagger(x) \\ &\simeq \mathbb{1} \otimes \mathbb{1} + \varepsilon Q \otimes \mathbb{1} + \varepsilon R \otimes \psi^\dagger(x) \\ &\simeq \exp\left[\varepsilon \left(Q \otimes \mathbb{1} + R \otimes \psi^\dagger(x) \right) \right] \end{aligned}$$

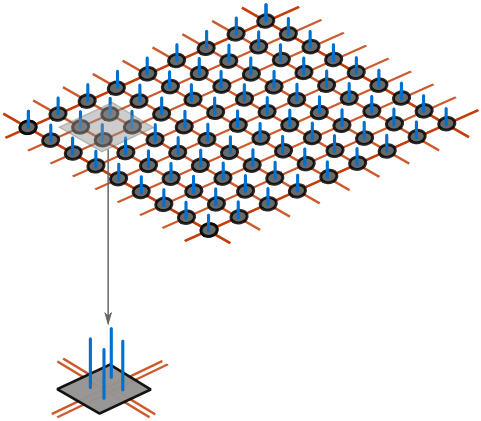
Computations

Thermodynamic limit

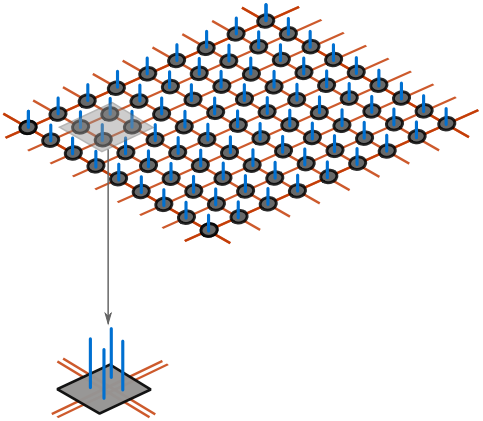
$$\langle \mathcal{O}(x) \mathcal{O}(y) \rangle = \mathrm{tr}\left(\Phi_{\mathcal{O}} \cdot e^{-(y-x)T} \Phi_{\mathcal{O}} \cdot \rho_{\mathrm{stat}} \right)$$

with $T = Q \otimes \mathbb{1} + \mathbb{1} \otimes \bar{Q} + R \otimes \bar{R}$

Continuous Tensor Networks: renormalization



Continuous Tensor Networks: renormalization



Upon blocking:

1. The physical dimension increases (idem cMPS $\Longrightarrow$ physical field)
2. The bond dimension increases too

we have to accept to lose some properties, especially finite bond dimension (at first).

Choice of trivial tensor

For **MPS**, not much choice:



Choice of trivial tensor

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For **TNS** in $d \geqslant 2$, many options:

1. Take a δ between all legs $\sim$ GHZ state $T^{(0)} =$ 
   $\Longrightarrow$ trivial geometry

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   $\Longrightarrow$ breakdown of Euclidean invariance

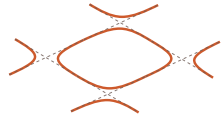
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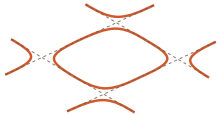
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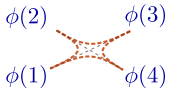
We will consider a softer modification of the first version:

$T^{(0)} \sim$ 

Ansatz

1 – Take a "Trivial" tensor:

$$T^{(0)}_{\phi(1),\phi(2),\phi(3),\phi(4)} =$$



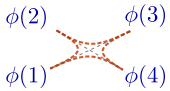
$$\sim \exp\left\{ \frac{-1}{2} \sum_{k=1}^{D} [\phi_k(1) - \phi_k(2)]^2 + [\phi_k(2) - \phi_k(3)]^2 + [\phi_k(3) - \phi_k(4)]^2 + [\phi_k(4) - \phi_k(1)]^2 \right\}$$

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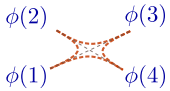
2 – And add a "correction":

$$\exp\left\{ -\varepsilon^2 V[\phi(1), \cdots, \phi(4)] + \varepsilon^2 \alpha[\phi(1), \cdots, \phi(4)] \psi^\dagger(x) \right\}$$

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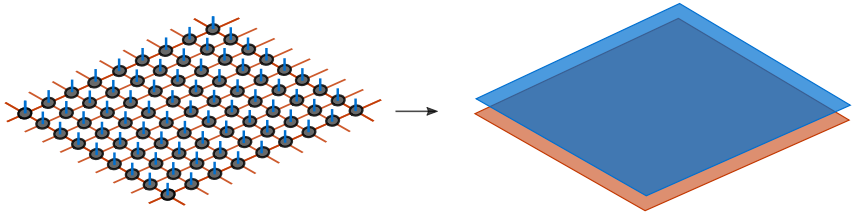
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3 – Realize tensor contraction = functional integral and trivial tensor gives free field measure.

Functional integral definition

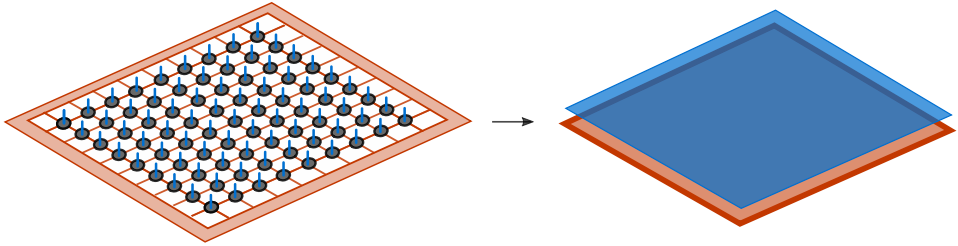
In the continuum limit:



$$|V,\alpha\rangle = \int \mathcal{D}\phi \, \exp\left\{-\int_\Omega \mathrm{d}^d x \, \frac{1}{2}\sum_{k=1}^{D} [\nabla\phi_k(x)]^2 + V[\phi(x)] - \alpha[\phi(x)]\,\psi^\dagger(x)\right\}|0\rangle$$

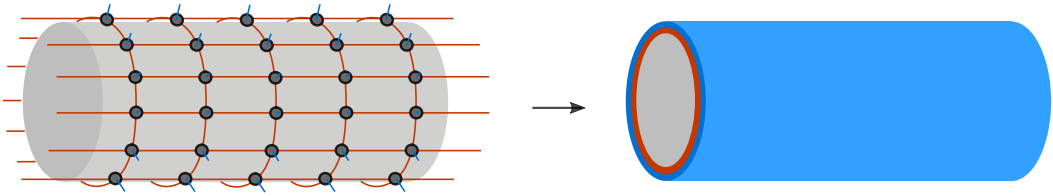
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Operator definition

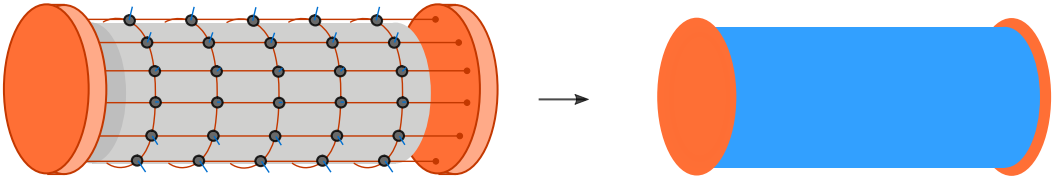


$$|V,\alpha\rangle = \mathrm{tr}\left[\mathcal{T}\exp\left(-\int_0^T d\tau \int_S dx\, \frac{\hat{\pi}_k(x)\hat{\pi}_k(x)}{2} + \frac{\nabla\hat{\phi}_k(x)\nabla\hat{\phi}_k(x)}{2} + V[\hat{\phi}(x)] - \alpha[\hat{\phi}(x)]\,\psi^\dagger(\tau,x)\right)\right]|0\rangle$$

where:

- $\hat{\phi}_k(x)$ and $\hat{\pi}_k(x)$ are k independent canonically conjugated pairs of (auxiliary) field *operators*: $[\hat{\phi}_k(x),\hat{\phi}_l(y)]=0$, $[\hat{\pi}(x)_k,\hat{\pi}_l(y)]=0$, and $[\hat{\phi}_k(x),\hat{\pi}_l(y)]=i\delta_{k,l}\,\delta(x-y)$ acting on a space of $d-1$ dimensions.

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Expressivity and stability

How big are cTNS?

Stability

The sum of two cTNS of bond field dimension D_1 and D_2 is a cTNS with bond field dimension $D \leqslant D_1 + D_2 + 1$:

$$|V_1, \alpha_1\rangle + |V_2, \alpha_2\rangle = |W, \beta\rangle$$

Expressivity

All states in the Fock space can be approximated by cTNS:

- A field coherent state is a cTNS with $D = 0$
- Stability allows to get all sums of field coherent states

Note: expressivity can also be obtained with $D = 1$ but it is less natural. Flexibility in D makes the expressivity higher for restricted classes of V and α.

Computations

Define generating functional for normal ordered correlation functions

$$\mathcal{Z}_{j',j} = \frac{1}{\langle V,\alpha|V,\alpha\rangle}\langle V,\alpha|\exp\left(\int \mathrm{d}x\, j'(x)\psi^\dagger(x)\right)\exp\left(\int \mathrm{d}x\, j(x)\psi(x)\right)|V,\alpha\rangle$$

Functional integral representation

- Use formula for overlap of field coherent states

$$\langle\beta|\alpha\rangle = \exp\left(\int \mathrm{d}x\, \beta^*(x)\,\alpha(x)\right)$$

- Compute with Gaussian integration + Feynman diagrams or Monte Carlo

Operator representation

Similar to cMPS

- Transfer matrix

$$\langle\mathcal{O}(x)\mathcal{O}(y)\rangle = \mathrm{tr}\left(\Phi_{\mathcal{O}}\cdot e^{-(y-x)T}\Phi_{\mathcal{O}}\cdot\rho_{\mathrm{stat}}\right)$$

with $T = Q\otimes\mathbb{1} + \mathbb{1}\otimes\bar{Q} + R\otimes\bar{R}$ with

$$Q = -\int \frac{\hat{\pi}_k(x)^2 + [\nabla\hat{\phi}_k(x)]^2}{2} + V(\hat{\phi}(x))$$

and $R\otimes\bar{R} = \int V(\hat{\phi}(x))\otimes V(\hat{\phi}(x))^\dagger$

Redundancies

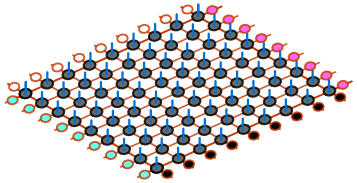
Discrete redundancy

Different elementary tensors are **equivalent**, they give the same state:



when  and 

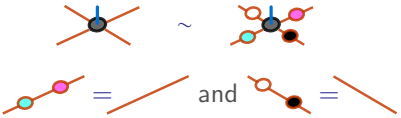
up to **boundary** terms:



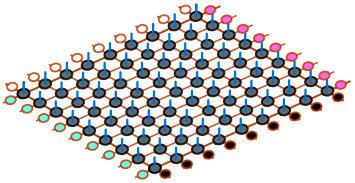
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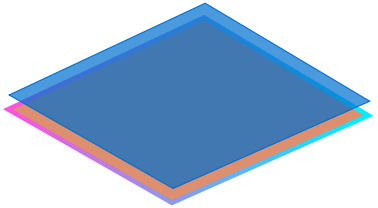


Continuum redundancy

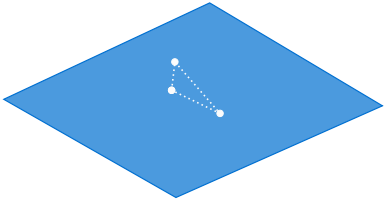
$$V(\phi) \to V(\phi) + \nabla \cdot \mathscr{F}[x, \phi(x)]$$

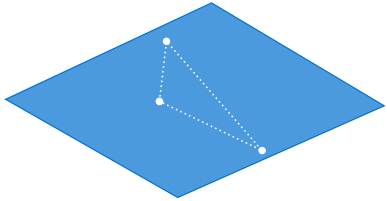
Just Stokes' theorem. If Ω has a boundary $\partial\Omega$:

$$\mathcal{D}[\phi] \to \mathcal{D}[\phi] \exp\left\{ \oint_{\partial\Omega} \mathrm{d}^{d-1}x \, \mathscr{F}[x, \phi(x)] \cdot \mathbf{n}(x) \right\}$$



Renormalization



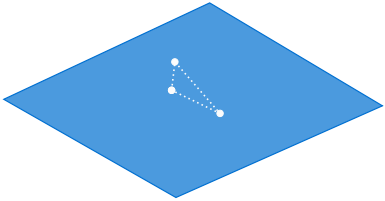


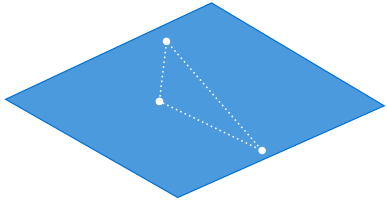
$$C(x_1, \cdots, x_n) = \langle T(1)|\mathcal{O}(x_1)\cdots\mathcal{O}(x_n)|T(1)\rangle,$$

the objective is to find a tensor $T(\lambda)$ of new parameters such that:

$$C(\lambda x_1, \cdots, \lambda x_n) \propto \langle T(\lambda)|\mathcal{O}(x_1)\cdots\mathcal{O}(x_n)|T(\lambda)\rangle.$$

Renormalization





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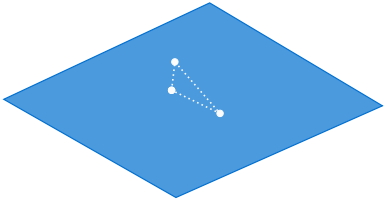
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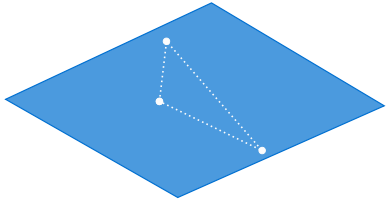
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Doable exactly:

$$V \to \lambda^d V \circ \lambda^{\frac{2-d}{2}} \quad \text{and} \quad \alpha \to \lambda^{\frac{d}{2}}\alpha \circ \lambda^{\frac{2-d}{2}}$$

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- $d = 2$, All powers of the field in V and α yield relevant couplings
- $d = 3$, The powers $p = 1, 2, 3, 4, 5$ of the field in V yield relevant $\Delta > 0$ couplings. The power $p = 6$ is marginal in V. For α, the powers $p = 1, 2$ are relevant and $p = 3$ is marginal. All other powers are irrelevant.

Getting back cMPS

One can get back cMPS with finite bond dimension by:

1. **Compactification** Take $d-1$ dimensions out of d to be very small



$$|V, B, \alpha\rangle \simeq \mathrm{tr}\left[\hat{B}\,\mathcal{T}\exp\left(-\int_0^T \mathrm{d}\tau \sum_{k=1}^{D} \frac{\hat{P}_k^2}{2} + V[\hat{X}] - \alpha[\hat{X}]\,\psi^\dagger(\tau)\right)\right]|0\rangle$$

$\Longrightarrow$ Hilbert space of a quantum particle in D space dimensions.

2. **Quantization** Take V with D deep minima to force the auxiliary field to take only D possibilities

Generalization

For a general Riemanian manifold $\mathcal{M}$ with boundary $\partial\mathcal{M}$, define:

$$|V, B, \alpha\rangle = \int \mathcal{D}\phi \, B(\phi|_{\partial\mathcal{M}}) \exp\left\{ -\int_{\mathcal{M}} d^d x \sqrt{g} \left(\frac{g^{\mu\nu} \partial_\mu \phi_k \partial_\nu \phi_k}{2} + V[\phi, \nabla\phi] - \alpha[\phi, \nabla\phi] \, \psi^\dagger \right) \right\} |0\rangle$$

i.e. add curvature and possible anisotropies in V and α

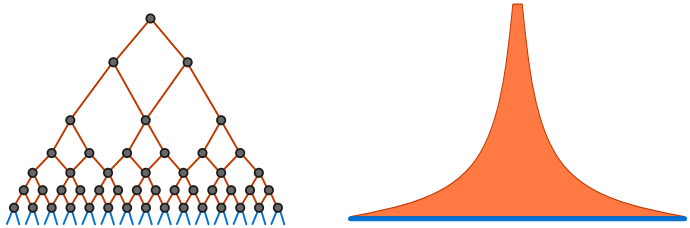
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Example: $\alpha[x, \phi, \nabla\phi]$ localized on the boundary and hyberbolic metrix g:



$\rightarrow$ **cMERA** in $d-1$ dimensions

Summary

$$|V, B, \alpha\rangle = \int \mathcal{D}\phi \, B(\phi|_{\partial\Omega}) \exp\left\{-\int_\Omega \mathrm{d}^d x \, \frac{1}{2} \sum_{k=1}^{D} [\nabla\phi_k(x)]^2 + V[\phi(x)] - \alpha[\phi(x)] \, \psi^\dagger(x)\right\} |0\rangle$$

Continuous tensor network states are natural continuum limits of tensor network states and natural higher d extensions of continuous matrix product states.

1. Obtained from discrete tensor networks
2. Can be made Euclidean invariant
3. Have functional and operator representations
4. Have a geometrical equivalent of the discrete gauge redundancies
5. Have an exact and explicit renormalization flow

